DATE	28-02-2026
TEAM ID	LTVIP2026TMID****
PROJECT NAME	HEMATOVISION: ADVANCED BLOOD CELL CLASSIFICATION USING TRANSFER LEARNING
MAXIMUM MARKS	4 MARKS

2.3 BRAINSTORMING :

Brainstorming is a crucial phase in the development of HematoVision: Advanced Blood Cell Classification Using Transfer Learning. This stage focuses on identifying key challenges in traditional blood cell analysis and generating innovative, technology-driven solutions to overcome them. The goal was to explore multiple approaches before selecting the most effective and feasible method.

Identified Problems:

Manual blood cell classification is time-consuming and requires continuous concentration.
Results may vary depending on the experience of the laboratory professional.
Increasing patient data leads to heavy workload in diagnostic laboratories.
Limited access to skilled hematologists in rural or resource-limited areas.
Developing deep learning models from scratch requires large datasets and high computational power.

Key Discussion Areas:

How can artificial intelligence improve medical image analysis?
What techniques can reduce training time without sacrificing accuracy?
How can the system ensure reliable and consistent results?
What evaluation metrics should be used to measure performance?
How can the solution be made user-friendly for laboratory integration?

Generated Ideas:

Build a deep learning-based automated blood cell classification system.
Use transfer learning with pre-trained CNN models such as ResNet, VGG, or MobileNet.
Apply image preprocessing techniques like resizing, normalization, and data augmentation.
Incorporate model validation techniques to prevent overfitting.
Create a simple graphical interface for easy usage in laboratories.
Enable the system to provide confidence scores along with predictions.